

INTERESTS	Quantum Computer Architecture, Software Stack, Error Correction and Detection	
EDUCATION	Sungkyunkwan University (SKKU) , Suwon, Korea	Mar 2020 – Feb 2026
	• B.S. in Electronic and Electrical Engineering • B.S. in Advanced Semiconductor Engineering – System Architecture Track • Cumulative GPA: 3.95 / 4.0 Major GPA: 4.0 / 4.0	
	The Pennsylvania State University , University Park, PA	Fall 2024
	• Exchange Student in Computer Engineering • Grade: 4.0 / 4.0	
PUBLICATIONS	[1] Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy SeokHyeon Kong, Donghwan Kim, Euseong Seo, and Kiwan Maeng CAL 2025 – IEEE Computer Architecture Letters	
	[2] Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system SeokHyeon Kong and Hayoung Oh JKIICE 2023 – Journal of the Korean Institute of Information and Communication Engineering	
POSTER PRESENTATION	[3] Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices SeokHyeon Kong, Chanhee Park, Jang-il Moon, and Denny Dahl Qcenter – Quantum Information Research Support Center, Korea	
RESEARCH EXPERIENCE	IonQ , Remote	Aug 2025 – Sep 2025
	<i>Research Mentee with Dr. Denny Dahl</i> • Addressed the Instant Insanity problem by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods [3]. • Solved the Four Corners Map Coloring problem using a Quantum Approximate Optimization Algorithm (QAOA) with two different encoding schemes [3].	
	The Pennsylvania State University , University Park, PA	Aug 2024 – Sep 2025
	<i>Undergraduate Researcher with Prof. Kiwan Maeng</i> • Conducted GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differential privacy [1]. • Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality [1].	
	Arch Lab at SKKU , Suwon, Korea	Jul 2025 – Aug 2025
	<i>Undergraduate Research Intern with Prof. Dongmoon Min</i> • Built foundational expertise in quantum / cryogenic computer systems through literature reviews, technical presentations and academic defenses. • Investigated quantum computer systems, including superconducting-qubit design, error mechanisms, control and readout techniques.	

	POSTECH QCQN Lab , Pohang, Korea <i>Undergraduate Research Intern</i>	Jan 2025
	- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^+ ions and enhanced system stability through targeted adjustments. - Visualized micromotion of trapped ions via 2D scans of DC compensation rods and minimized unwanted oscillatory motion.	
TECHNICAL PROJECT	SoC Design and Practice , Sungkyunkwan University (SKKU) <i>Graduate Level Coursework</i>	Fall 2025
	- Designed and implemented a multi-channel DMAC engine in SystemVerilog, integrating AMBA APB for configuration and AXI for data transfer. - Synthesized and verified the design through area and timing analysis, optimizing performance to meet target specifications.	
TEACHING EXPERIENCE	Academic Tutor , Sungkyunkwan University (SKKU) <i>Intro to Electromagnetism</i>	Fall 2023
	Led weekly Q&A sessions and problem-solving discussions.	
	Teaching Assistant , Sungkyunkwan University (SKKU) <i>Engineering Mathematics 2</i>	Summer 2023
	Prepared and organized class materials, including exams.	
HONORS AND AWARDS	Director's Award – Quantum Hackathon, QCenter President's List , SKKU Dean's List , SKKU Korea–U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000), KIAT Student Success Scholarship , SKKU	2025 2024, 2025 SP2023, FA2023, SP2024, FA2024, SP2025 FA2024 FA2023
SKILLS	Software: Qiskit, CUDA, C, Python, MATLAB, LaTeX Hardware: System Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute	
LEADERSHIP AND SERVICE	Executive Board , President's List (Honor Society at SKKU) Qiskit Advocate , IBM Quantum Squad Leader , Republic of Korea Army (Military Service)	2025 2025 Apr 2021 – Oct 2022
REFERENCES	Kiwan Maeng, Ph.D. Assistant Professor of Computer Science and Engineering The Pennsylvania State University Denny Dahl, Ph.D. Customer Success Director IonQ Bong-Shik Song, Ph.D. Professor of Electronic and Electrical Engineering Sungkyunkwan University (SKKU)	Email: kvm6242@psu.edu Email: denny.dahl@ionq.co Email: songwiz@skku.edu